

EE 6271- BASIC ELETRONICS

LECTURE 1a:

SEMICONDUCTOR DIODES AND APPLICATIONS

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ELECTRONIC DEVICES VS ELECTRICAL

- Comparing these two types of engineering is like comparing the meaning of electric vs electronic.
- **Electrical devices** convert electrical energy into other forms of energy, for example heat, light or sound.
- **Electronic devices** process and control the flow of electrons (electrical signals) in order to perform a task such as information processing, storage and transmission,
- While **electrical engineering** is concerned with the large-scale production and distribution of electrical power, **electronic engineering** focuses on smaller-scale electrical circuits and flow of electrons to process information or signals.

ELECTRONIC DEVICES VS ELECTRICAL

- **Electronics** may be defined as the science and technology of electronic devices and systems.
- Electronic devices are primarily non-linear devices such as diodes and transistors and in general integrated circuits (ICs) in which small signals (voltages and currents) are applied to them.
- Resistors, capacitors and inductors existed long ago before the advent of semiconductor diodes and transistors, these devices are thought of as electrical devices and the systems that consist of these devices are generally said to be electrical rather than electronic systems.

APPLICATION OF ELECTRONICS

- Some of its applications included office gadgets like computers, scanners, calculators, FAX machines, projectors etc.
- It also includes home appliances like washing machines, refrigerators, microwaves, TVs, vacuum cleaners, video games, loudspeakers etc. and some advanced storage devices such as HDD jukebox, DVDs etc.

MATERIALS USED IN ELECTRONICS

- In terms of their electrical properties, materials can be classified into three groups: **conductors**, **semiconductors**, and **insulators**.
 - ❖ **An insulator** is a material that does not conduct electrical current under normal conditions. Eg. Wood, paper
 - ❖ **A conductor** is a material that easily conducts electrical current. Eg metals, salt, water
 - ❖ **A semiconductor** is a material that is between conductors and insulators in its ability to conduct electrical current

WHAT IS ELECTRICITY?

- ❖ All matter is composed of atoms; all atoms consist of electrons, protons, and neutrons except normal hydrogen, which does not have a neutron.
- ❖ Electrons (- charge) are attracted to protons (+ charge), this holds the atom together
- ❖ Some materials have strong attraction and refuse to loss electrons, these are called **insulators** (air, glass, rubber, most plastics)
- ❖ Some materials have weak attractions and allow electrons to be lost, these are called **conductors** (copper, silver, gold, aluminum)
- ❖ Electrons can be made to move from one atom to another, this is called a **current of electricity**

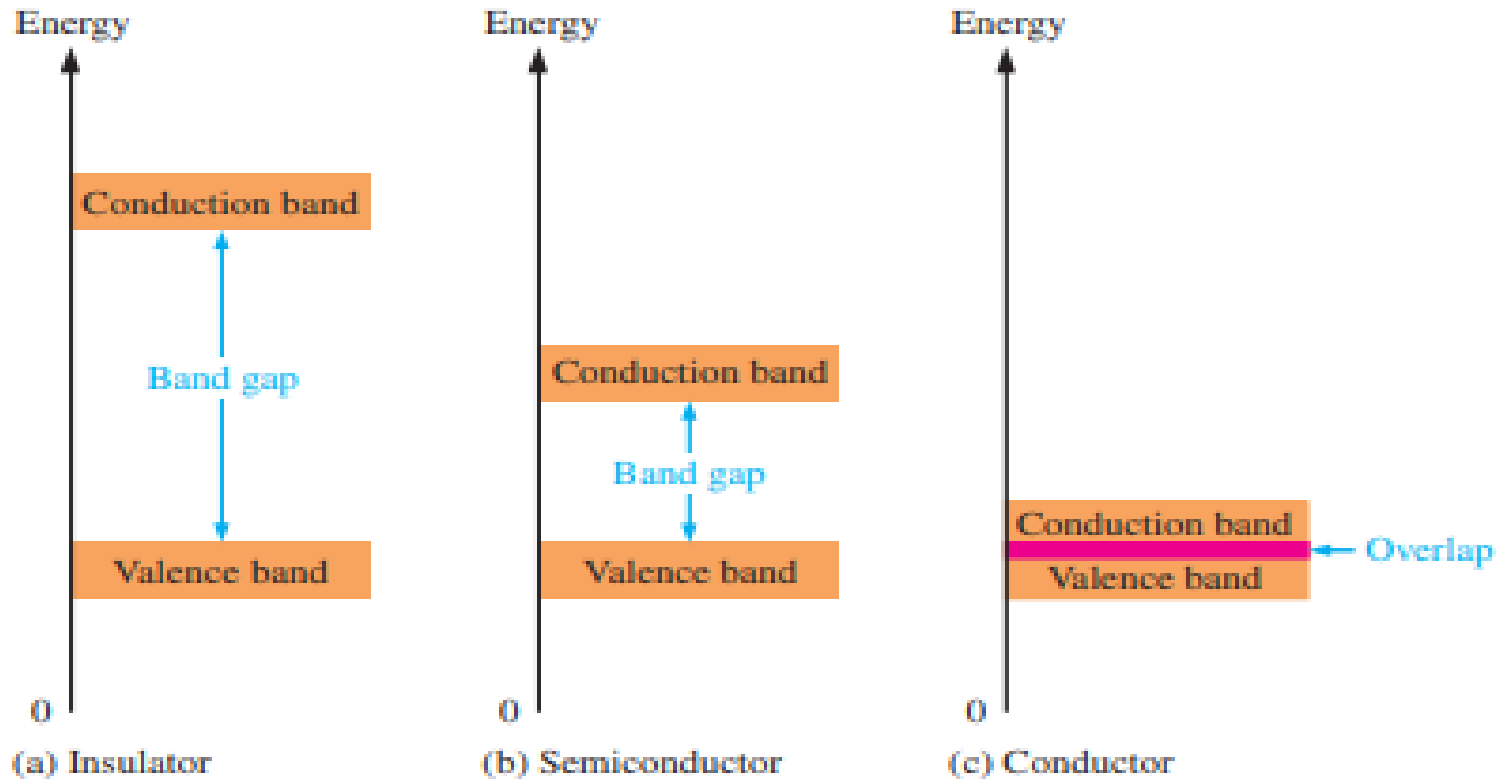
SHELLS, ORBITS, AND ENERGY LEVELS

- In the Bohr model, electrons orbit the nucleus of an atom at certain distances from the nucleus and are restricted to these specific orbits.
- The **nucleus** consists of positively charged particles called **protons** and uncharged particles called **neutrons**. The basic particles of negative charge are called **electrons**
- Each orbit corresponds to a different energy level within the atom known as a **shell**.
- The shells are designated 1, 2, 3, and so on, with 1 being closest to the nucleus. Electrons further from the nucleus are at higher **energy levels**.

BAND GAP

- ❑ Recall, that the valence shell of an atom represents a band of energy levels and that the valence electrons are confined to that band.
- ❑ When an electron acquires enough additional energy, it can leave the valence shell, become a free electron, and exist in what is known as the **conduction band**
- The difference in energy between the valence band and the conduction band is called an **energy gap** or **band gap**.
- This is the amount of energy that a valence electron must have in order to jump from the valence band to the conduction band.

BAND GAP

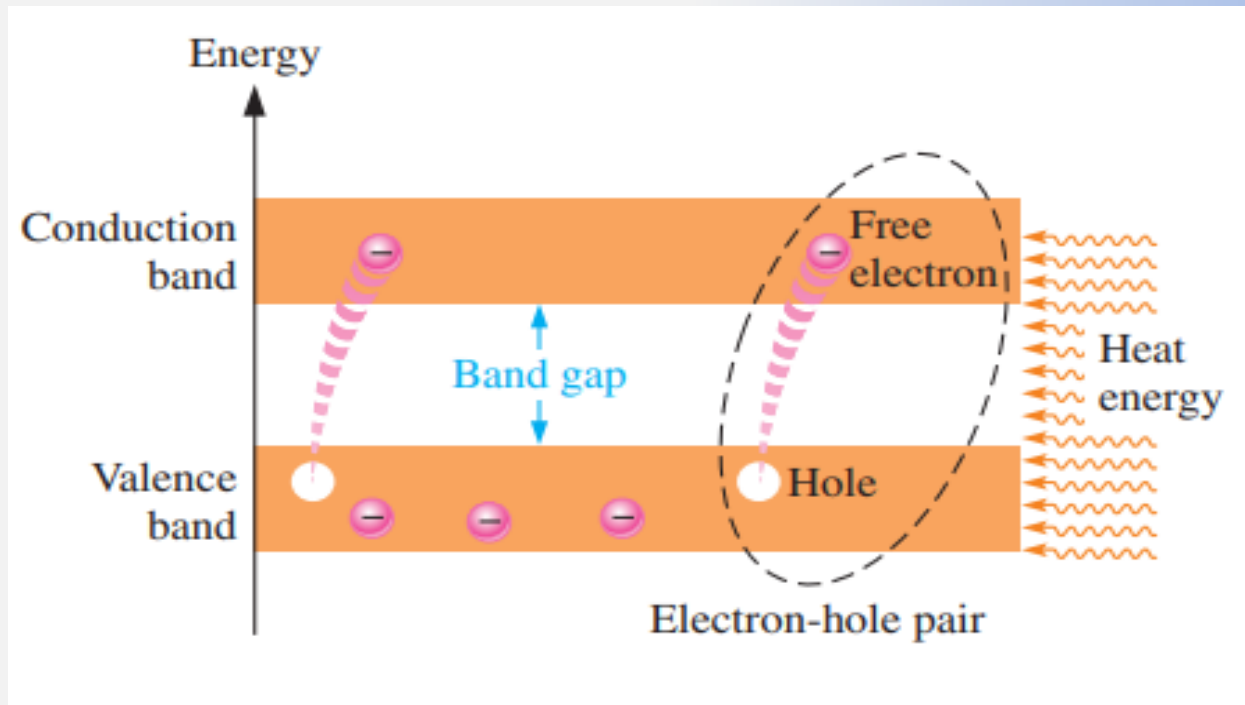


CONDUCTION IN SEMICONDUCTORS

- A **semiconductor** is a material that is between conductors and insulators in its ability to conduct electrical current.
- A semiconductor in its pure (intrinsic) state is neither a good conductor nor a good insulator.
- Single-element semiconductors are antimony (Sb), arsenic (As), astatine (At), boron (B), polonium (Po), tellurium (Te), silicon (Si), and germanium (Ge)
- Silicon and germanium are used in diodes, transistors, integrated circuits, and other semiconductor devices.

CONDUCTION ELECTRONS AND HOLES

- ❑ An intrinsic (pure) silicon crystal at room temperature has sufficient heat (thermal) energy for some valence electrons to jump the gap from the valence band into the conduction band, becoming free electrons.
- ❑ Free electrons are also called **conduction electrons**.

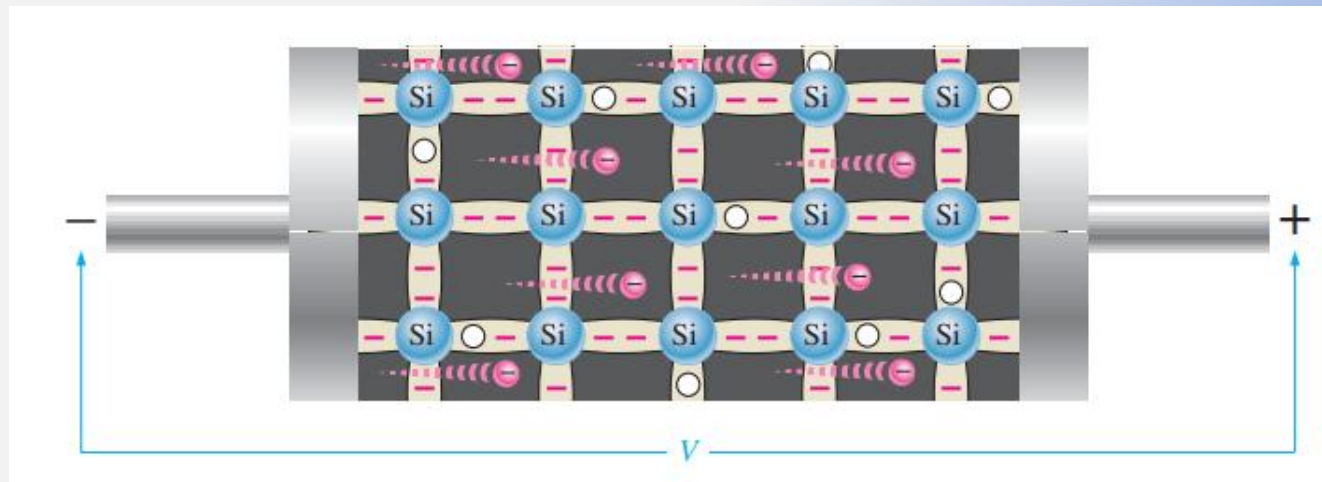


CONDUCTION ELECTRONS AND HOLES

- When an electron jumps to the conduction band, a vacancy is left in the valence band within the crystal. This vacancy is called a **hole**.
- For every electron raised to the conduction band by external energy, there is one hole left in the valence band, creating what is called an **electron-hole pair**.
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- **Recombination** occurs when a conduction-band electron loses energy and falls back into a hole in the valence band.

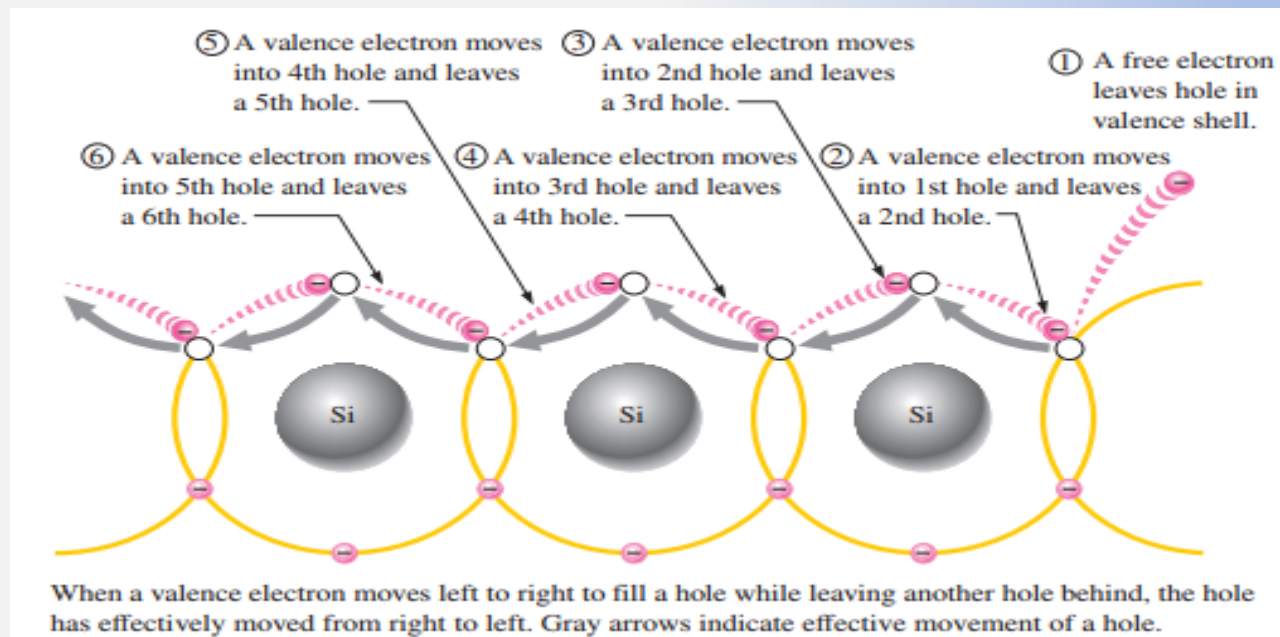
ELECTRON AND HOLE CURRENT

- When a voltage is applied across a piece of intrinsic silicon, the thermally generated free electrons in the conduction band, which are free to move randomly in the crystal structure, are now easily attracted toward the positive end.
- This movement of free electrons is one type of current in a semiconductive material and is called **electron current**.
- **Electron current** is the movement of the free thermally generated electrons in the conduction band when the voltage is applied in a semiconductor.



ELECTRON AND HOLE CURRENT

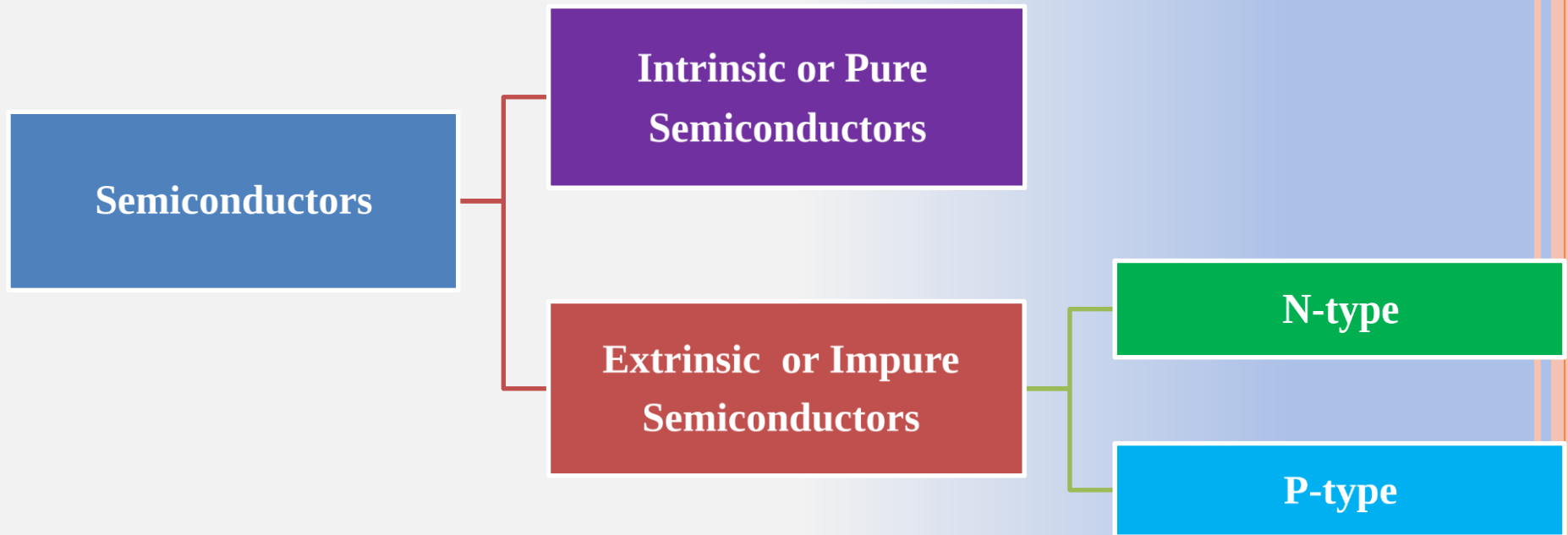
- **Hole current** is the movement of a valence electron into a nearby hole with little change in its energy level, thus leaving another hole where it came from.
- Although current in the valence band is produced by valence electrons, it is called hole current to distinguish it from electron current in the conduction band.



N-TYPE AND P-TYPE SEMICONDUCTORS

- Semiconductive materials do not conduct current well and are of limited value in their intrinsic state.
 - This is **because** of the limited number of free electrons in the conduction band and holes in the valence band.
- Intrinsic silicon (or germanium) must be modified by increasing the number of free electrons or holes to increase its conductivity and make it useful in electronic devices.
 - This is done by adding impurities to the intrinsic material.
 - This process, called **doping**, increases the number of current carriers (electrons or holes).
- **Doping** is the process of imparting impurities to an intrinsic semiconductive material in order to control its conduction characteristics.
- The two categories of impurities are **n-type** and **p-type**.

- ❑ Semiconductor may be classified as under:

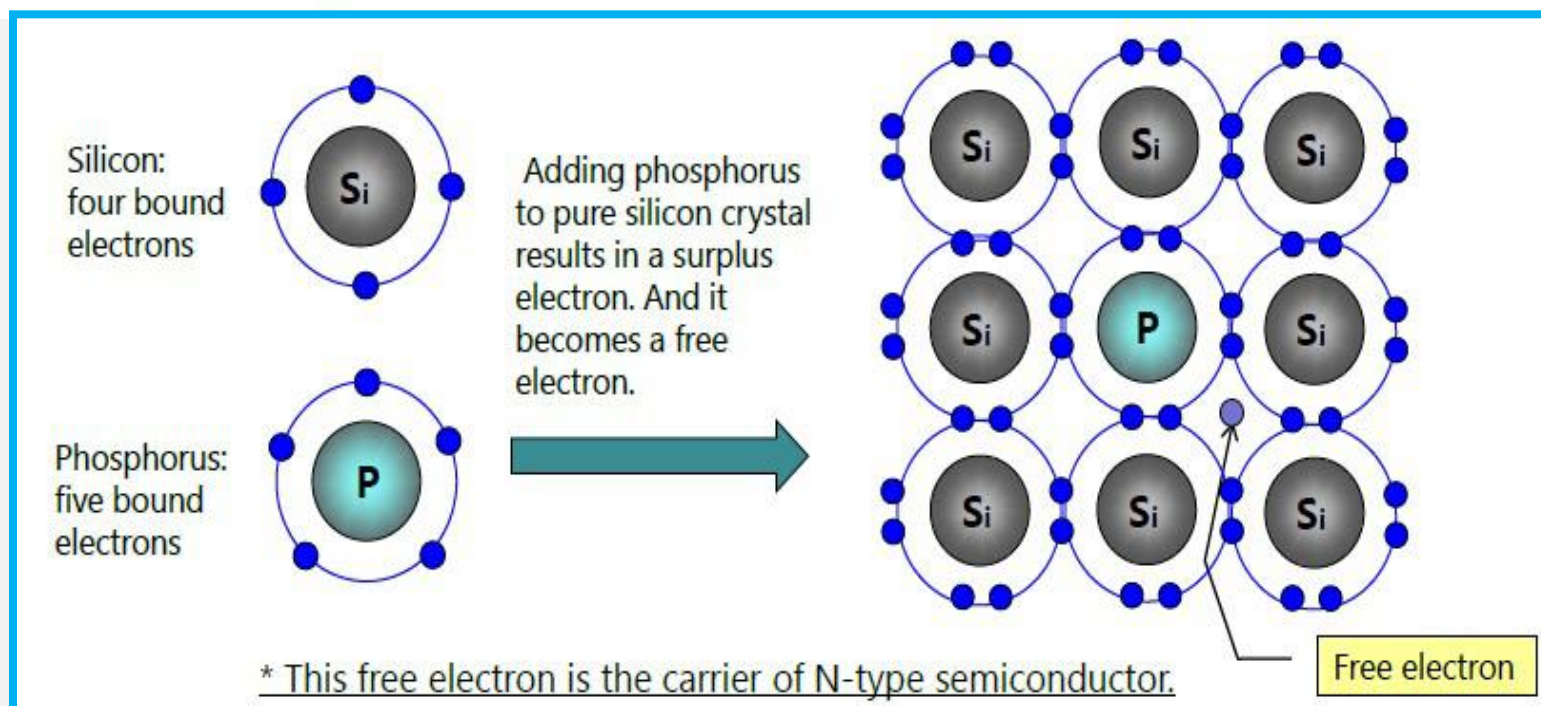


What is an N-type Semiconductor?

It is an intrinsic semiconductor doped with phosphorus (P), arsenic (As), etc.

Silicon of Group IV has four bound electrons, and phosphorus of Group V has five bound electrons.

An electron derived from the bound electrons of phosphorus can be made free (free electron) by adding a small amount of phosphorus to pure silicon crystal. Movement of the free electrons pulled to "+" electrode makes current flow.



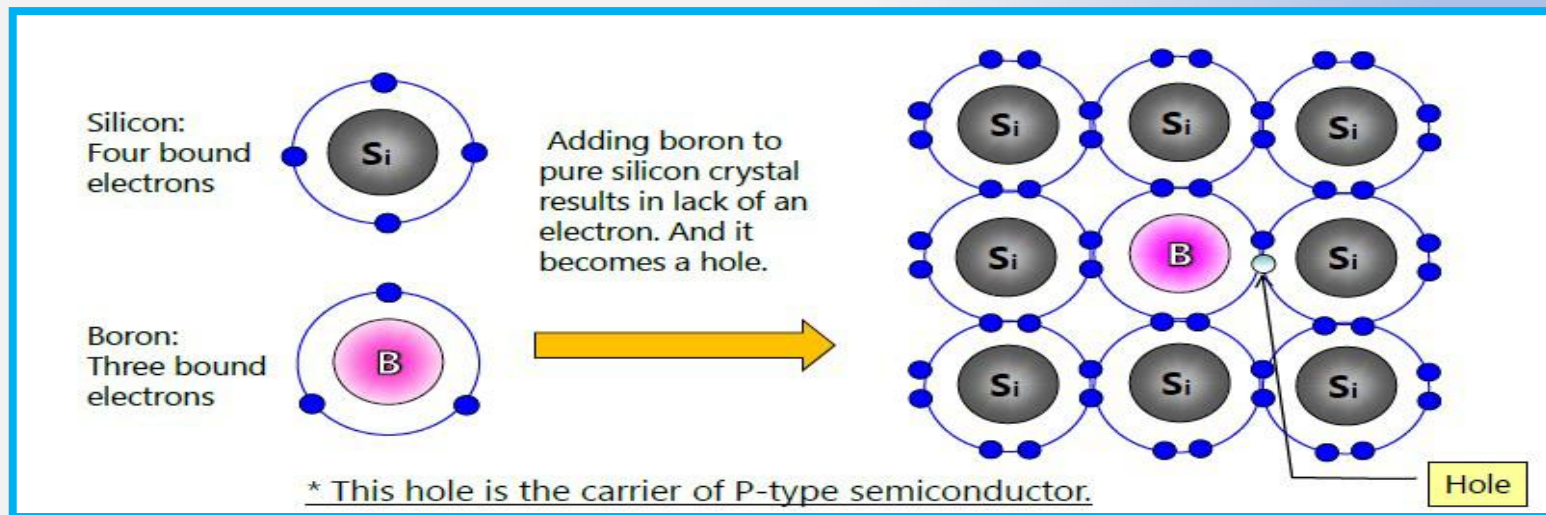
N-TYPE SEMICONDUCTOR

- To increase the number of conduction-band electrons in intrinsic silicon, **pentavalent** impurity atoms are added.
- These are atoms with five valence electrons such as
 - Arsenic (As),
 - Phosphorus (P),
 - Bismuth (Bi), and
 - Antimony (Sb).
- The pentavalent atom gives up an electron, it is often called a **donor atom**
- **Majority and Minority Carriers,**
 - The **electrons** are called the majority carriers in n-type material.
 - **Holes** in an n-type material are called minority carriers.

What is a P-type Semiconductor?

It is an intrinsic semiconductor doped with boron (B), indium (In), etc. Silicon of Group IV has four bound electrons, and boron of Group III has three bound electrons.

Where silicon and boron are bonded, lack of an electron can be brought about by adding a small amount of boron to pure silicon. This lack is called a hole. When voltage is applied in this situation, an adjacent electron moves to this hole and the place the electron is removed from becomes a new hole. Holes move to "-" electrode in sequence.



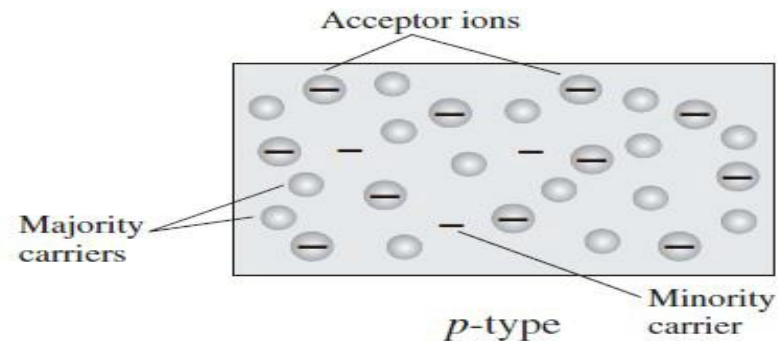
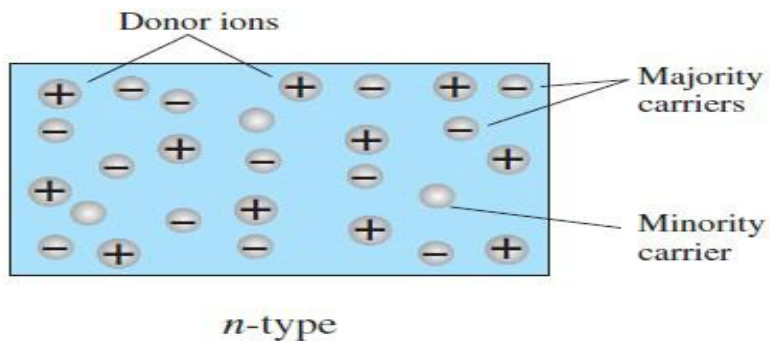
- 0 Difference between electrons and holes : holes unlike an electron which has a negative charge, **holes have a positive charge that is equal in magnitude but opposite in polarity to the charge an electron has.**

P-TYPE SEMICONDUCTOR

- To increase the number of holes in intrinsic silicon, **trivalent impurity** atoms are added.
 - These are atoms with three valence electrons such as
 - Boron (B),
 - Indium (In), and
 - Gallium (Ga).
 - The trivalent atom can take an electron; it is often referred to as an **acceptor atom**.
- **Majority and Minority Carriers,**
- The **holes** are called the **majority carriers** in p-type material.
 - Electrons in an p-type material are called **minority carriers**.

DIFFERENCE BETWEEN N-TYPE AND P-TYPE SEMICONDUCTOR

P-type material	N-type material
0 Semiconductor material doped with acceptors .	0 Semiconductor material doped with donors
0 Material has high hole concentration	0 Material has high concentration of free electrons .
0 Concentration of free electrons in p-type material is very low .	0 Concentration of holes in n-type material is very low .
0 Contains negatively charged acceptors (immovable) and positively charged holes (free).	0 Contains positively charged donors (immovable) and negatively charged free electrons .
0 Total charge = 0	0 Total charge = 0



THE PN JUNCTION

- When you take a block of silicon and dope part of it with a trivalent impurity and the other part with a pentavalent impurity, a **boundary** called the **pn junction** is formed between the resulting p-type and n-type portions.
- The pn junction is the basis for diodes, certain transistors, solar cells, and other devices.
- If a piece of intrinsic silicon is doped so that part is n-type and the other part is p-type, a pn junction forms at the boundary between the two regions and **a diode is created.**

THE PN JUNCTION

- The p region has many holes (majority carriers) from the impurity atoms and only a few thermally generated free electrons (minority carriers).
- The n region has many free electrons (majority carriers) from the impurity atoms and only a few thermally generated holes (minority carriers).

Semiconductor Diodes and Applications

□ What is a Diode ? Basic, Types, Characteristics, and Applications



Semiconductor Diodes and Applications

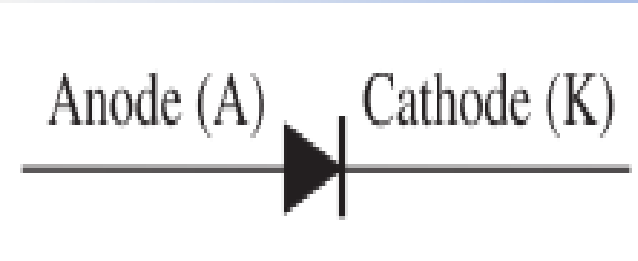
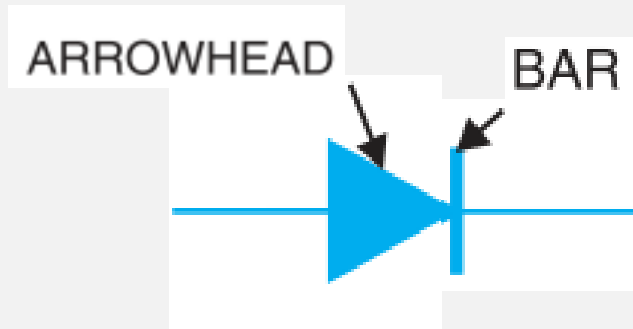
□ Types of diodes

- | | |
|--|---|
| <ul style="list-style-type: none">○ Small Current Diode○ Large Signal or Large Current Diodes○ Zener Diodes○ LED (Light Emitting Diodes)○ Photodiodes○ Constant Current Diodes○ Schottky Diode○ Shockley Diode○ Step Recovery Diodes○ Tunnel Diodes○ Varactor Diodes | <ul style="list-style-type: none">○ PIN Diodes○ LASER Diode○ Transient Voltage Suppression Diodes○ Gold Doped Diodes○ Super Barrier Diodes○ Point Contact Diodes○ Peltier Diodes○ Gunn Diode○ Vacuum Diodes○ Avalanche Diode○ Crystal Diode |
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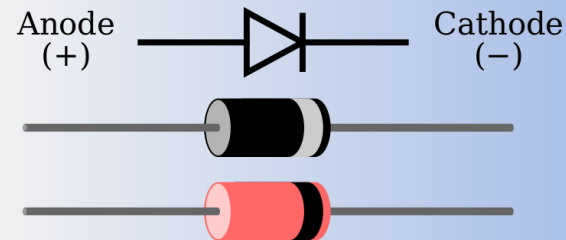
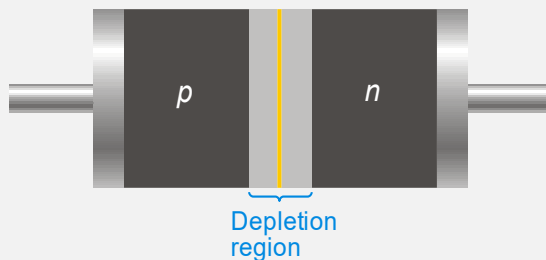


DIODE INTRODUCTION

- ❑ **Diode** is a two-terminal electronic component that conducts electric current in only one direction.



- ❑ The arrow in the symbol indicates the direction of easier conventional current flow.



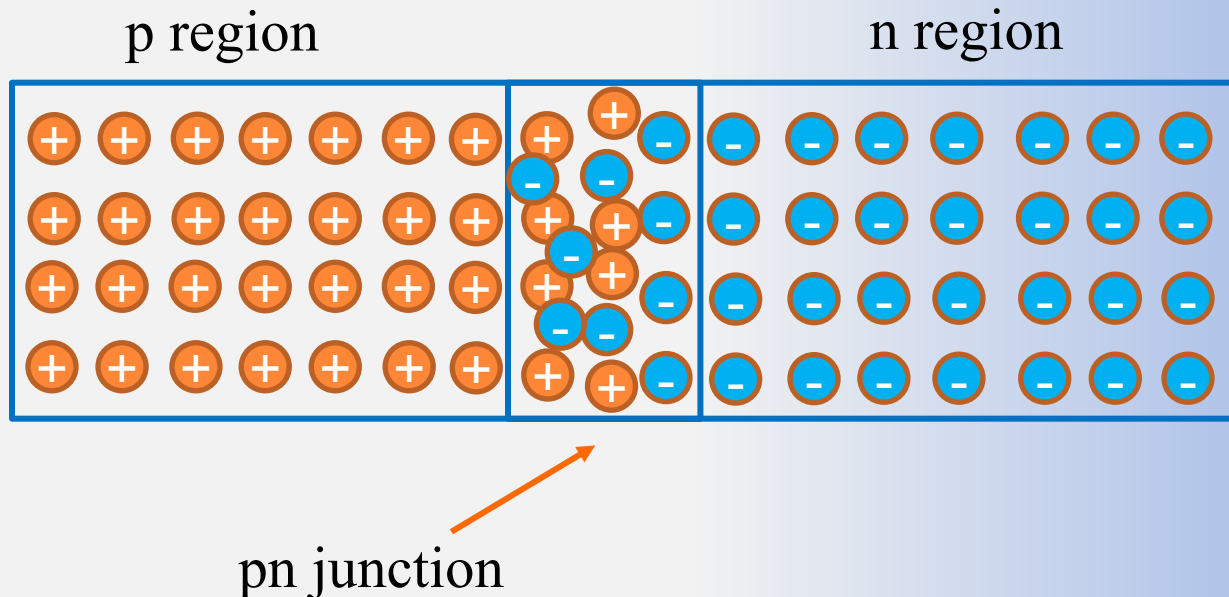
- ❑ A crystal diode has two terminals i.e Cathode and Anode.
- ❑ The p region is called **anode** and n type region is called **cathode**.
- ❑ It conducts current in one direction and offers high (ideally infinite) resistance in other direction.

FORMATION OF DEPLETION LAYER

❑ Diffusion:

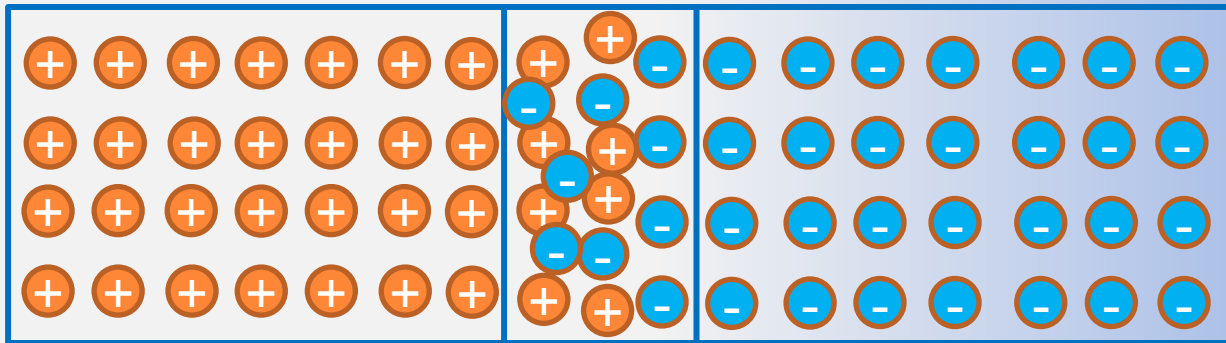
- Movement of electronics from a region of high concentration to lower concentration

- ## ❑ At the junction, the electrons from the n-type and holes from the p-type attract each other, combine and cancel their net charges.



FORMATION OF DEPLETION LAYER

- ❑ Due to the cancellation of electrons and holes at the junction, the junction will have no more carriers so it is called a **depleted region** or **depletion region**



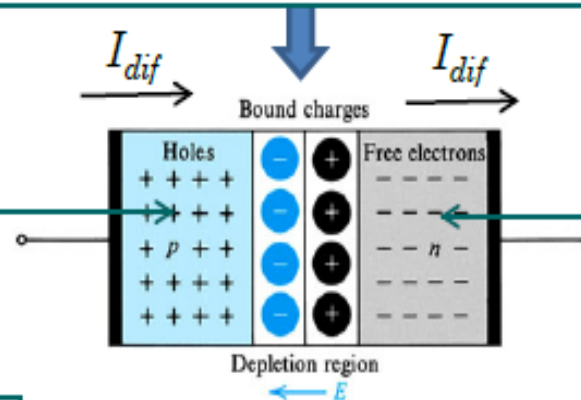
Depletion region



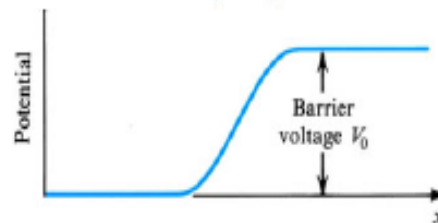
High concentration of h on the p side
Holes diffuse towards the junction

High concentration of e on the n side
Electrons diffuse towards the junction

Holes from the p side and electrons from the n side combine at the junction, forming a depletion region



p side is negatively charged because it has lost holes.



n side is positively charged because it has lost electrons.

A potential is formed which inhibits further diffusion of electron and holes (called junction built-in voltage)

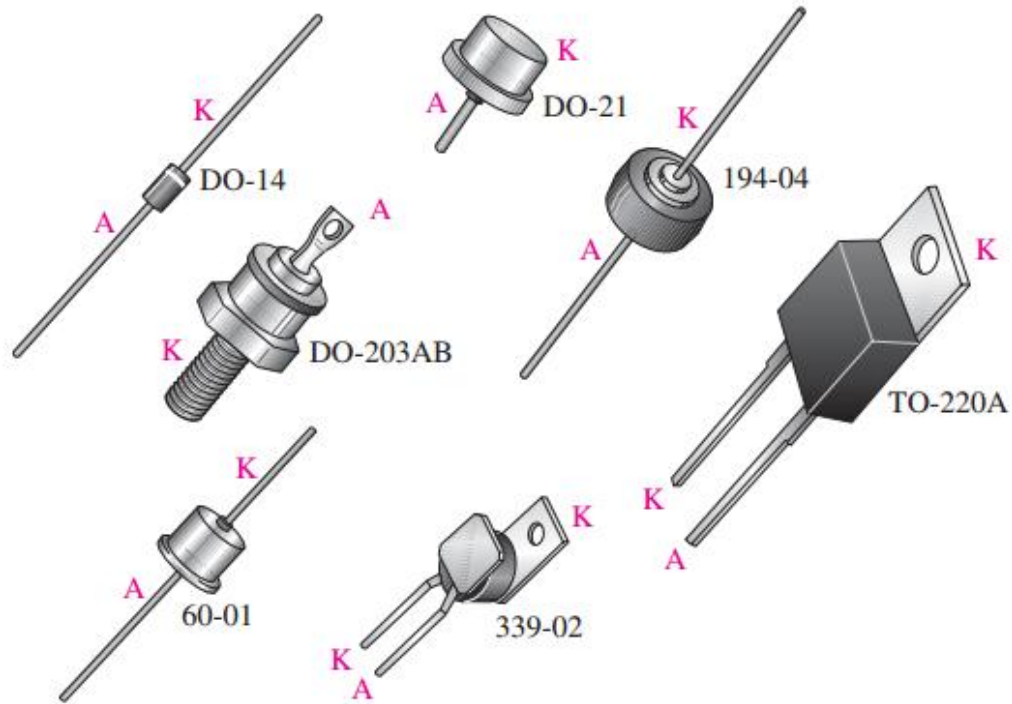
BARRIER POTENTIAL

- Any time there is a positive charge and a negative charge near each other, there is a force acting on the charges as described by Coulomb's law.
- In the depletion region there are many positive charges and many negative charges on opposite sides of the pn junction.
- The forces between the opposite charges form an **electric field**
- **Barrier potential** is the amount of voltage required to move electrons through the electric field of the pn junction.

BARRIER POTENTIAL

- The barrier potential of a pn junction depends on several factors, including the type of semiconductive material, the amount of doping, and the temperature.
- The typical barrier potential is approximately **0.7 V for silicon** and **0.3 V for germanium** at 25°C

TYPICAL DIODE PACKAGES



(a)



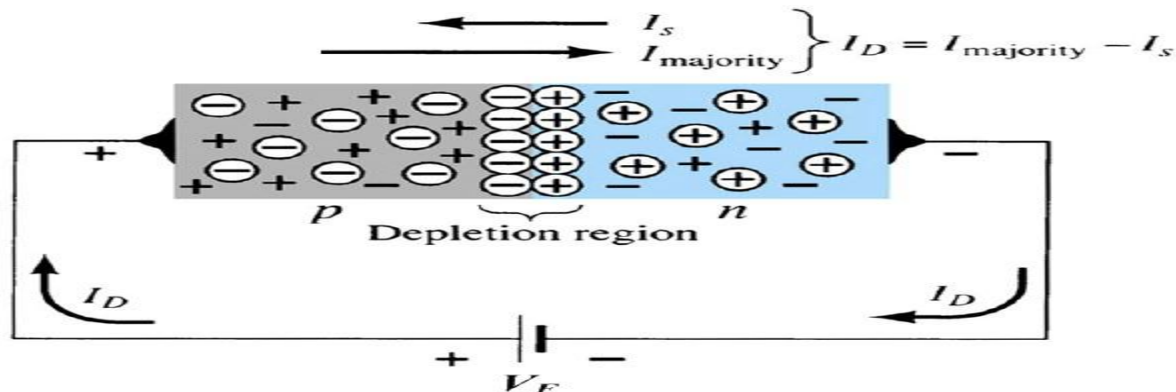
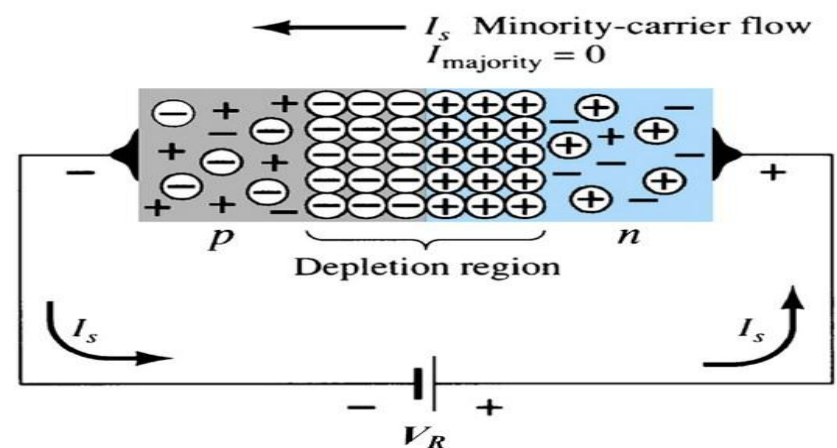
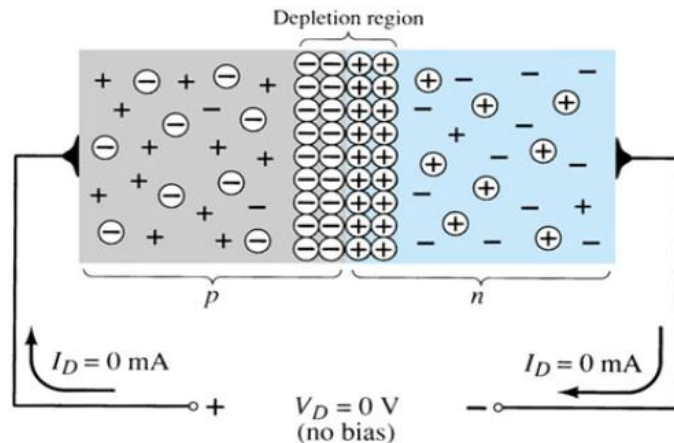
(b)



DIODE CONNECTION

□ Operating Conditions

- **No Bias:** No voltage is applied and no current is owing.
- **Reverse Bias:** Negative voltage (i.e., opposite polarity with the p-n junction) is applied.
- **Forward Bias:** Positive voltage (i.e., same polarity with the p-n junction) is applied.

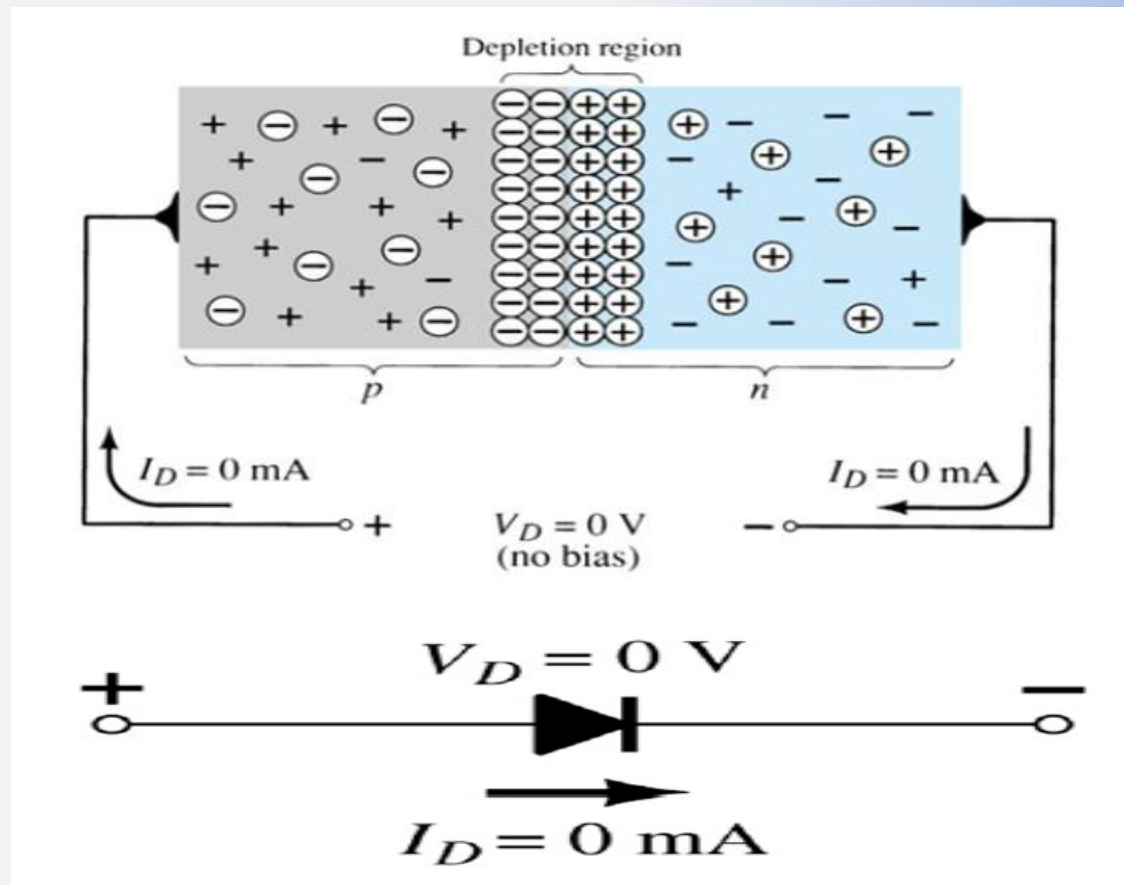


DIODE CONNECTION

❑ Operating Conditions : No Bias Condition

○ No Bias Condition

No external voltage is applied to the p-n junction as shown in Figure below. So, and no current is owing . Under no bias, only a modest depletion layer exists as seen in Figure below.

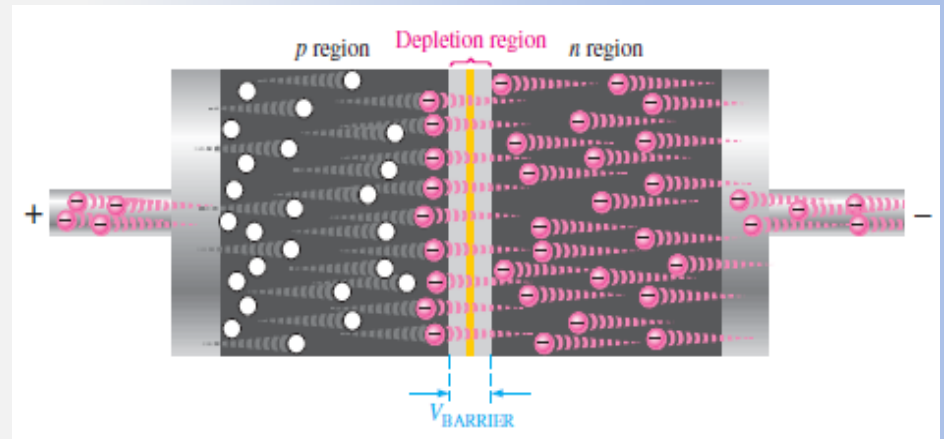


FORWARD BIASED

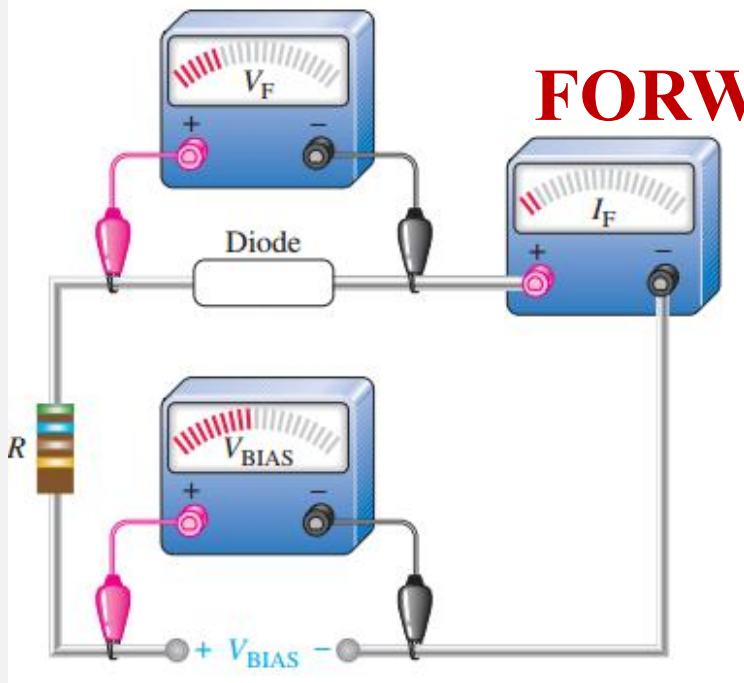
- Forward bias is a condition that allows current through pn junction.
 - ❑ A dc voltage (V_{Bias}) is applied to bias a diode.
 - ❑ **Positive side** is connected to **p-region** (anode) and **negative side** is connected with **n-region**.
 - ❑ V_{Bias} must be greater than 'barrier potential'

- ❑ As more electrons flow into the depletion region reducing the number of positive ions and similarly

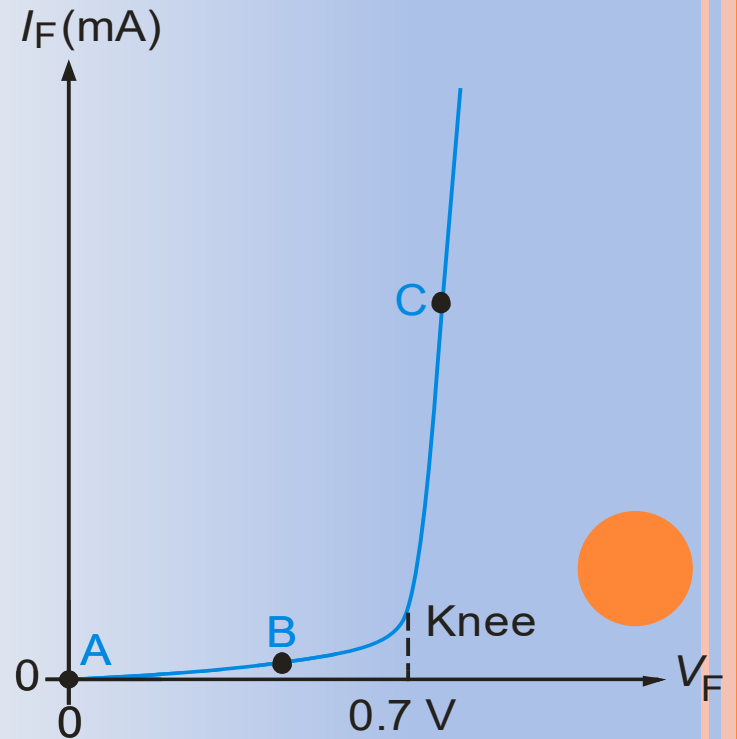
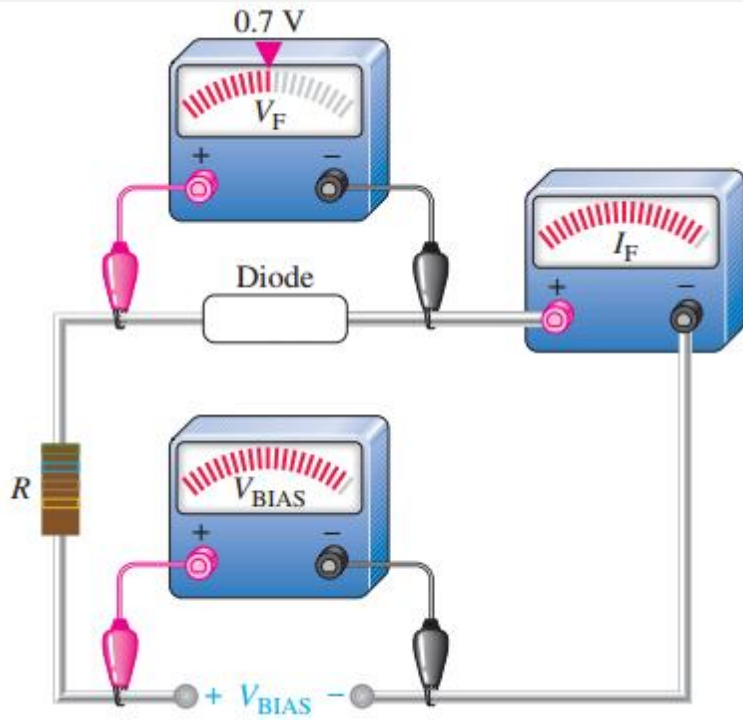
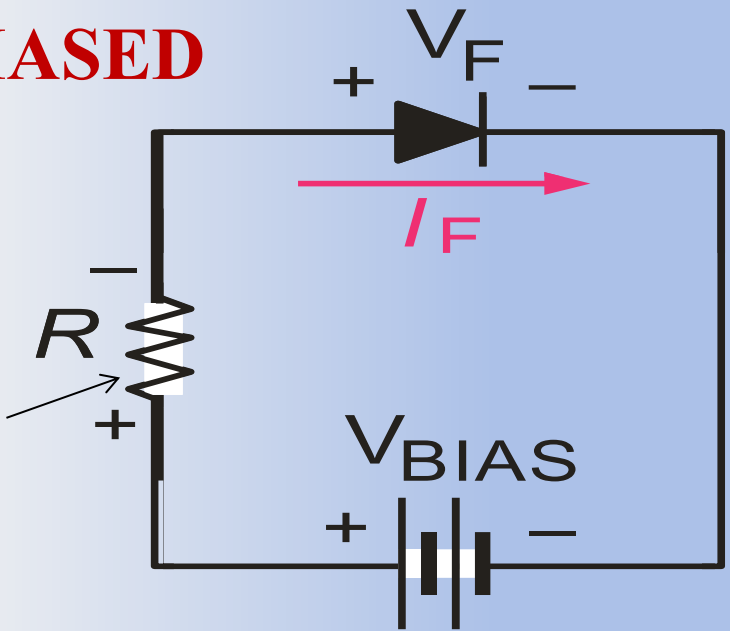
more holes move in reducing the positive ions. This reduces the width of depletion region.



FORWARD BIASED

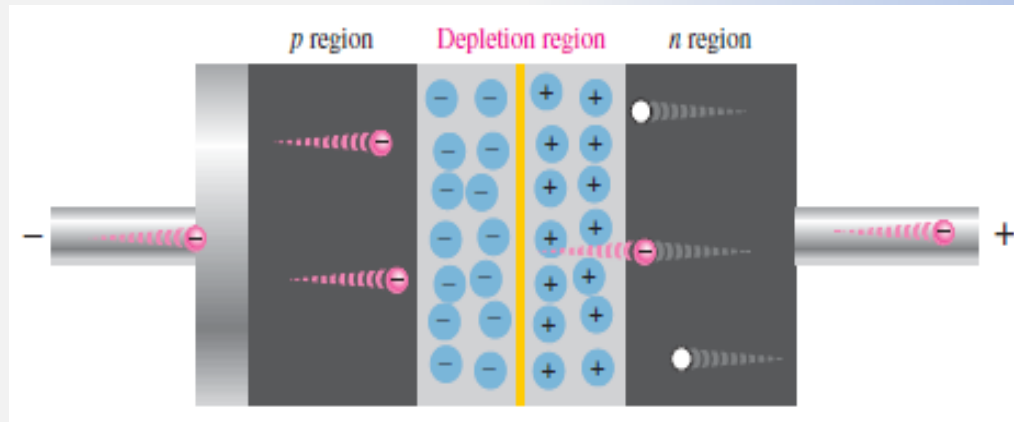


Current limiting resistor



REVERSE BIASED

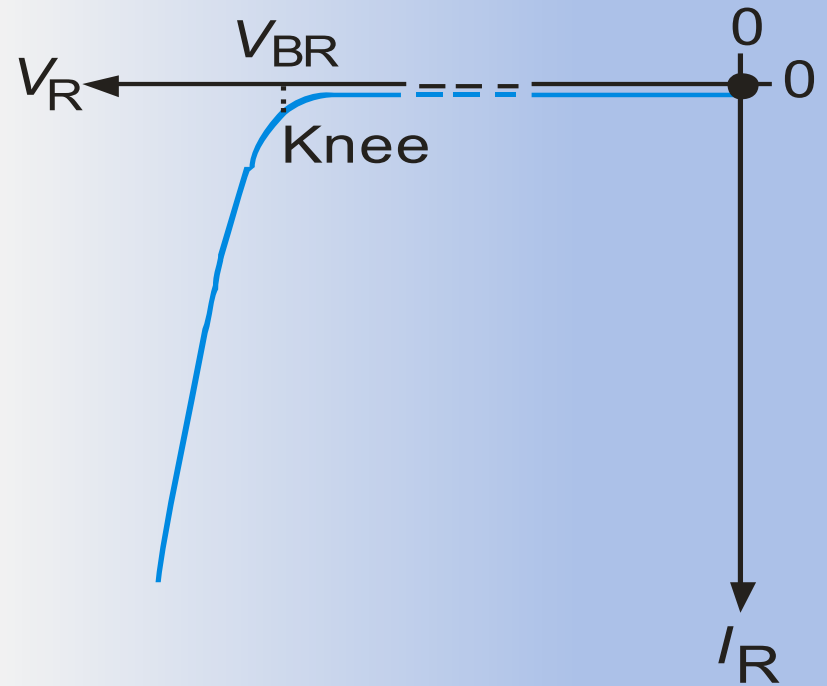
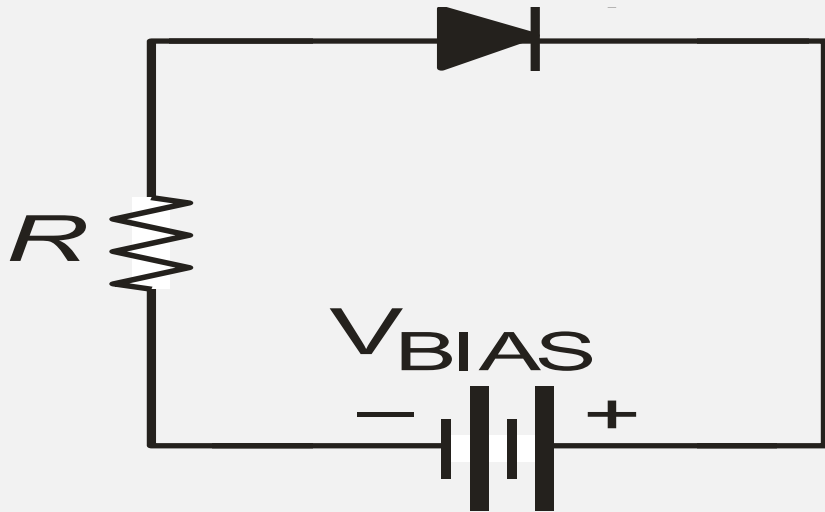
- ❑ Reverse bias is a condition that prevents current through junction.
- ❑ **Positive side** of V_{Bias} is connected to the **n-region** whereas the **negative side** is connected with **p-region**.
- ❑ Depletion region get wider with this configuration.



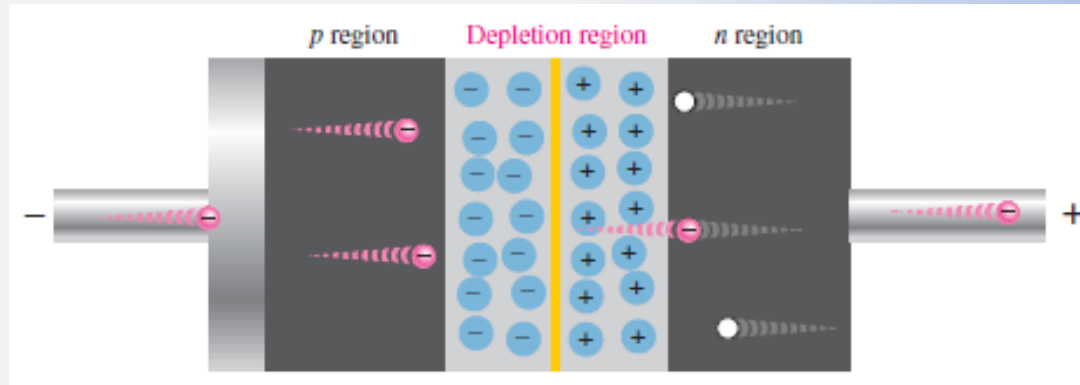
- ❑ The positive side of bias voltage attracts the majority carriers of n-type creating more positive ions at the junction.
- ❑ This widens the depletion region



REVERSE BIASED



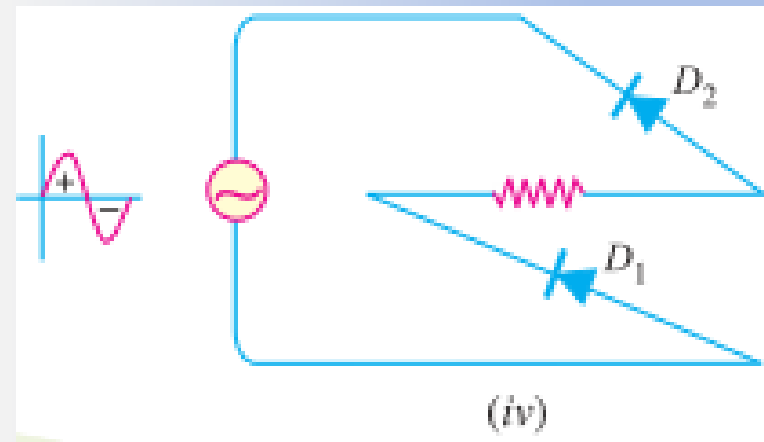
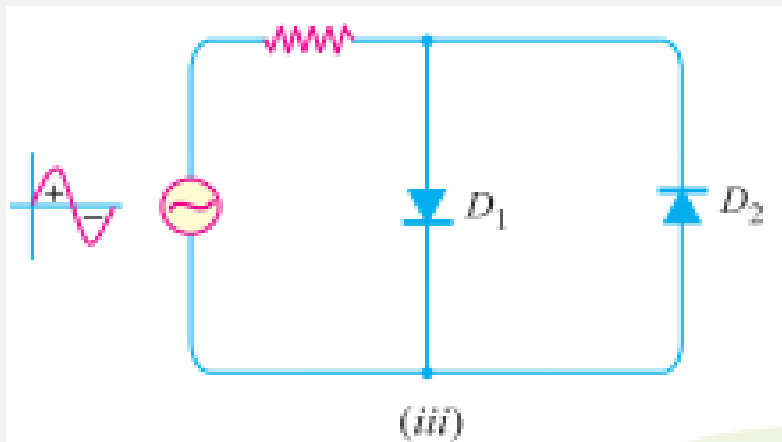
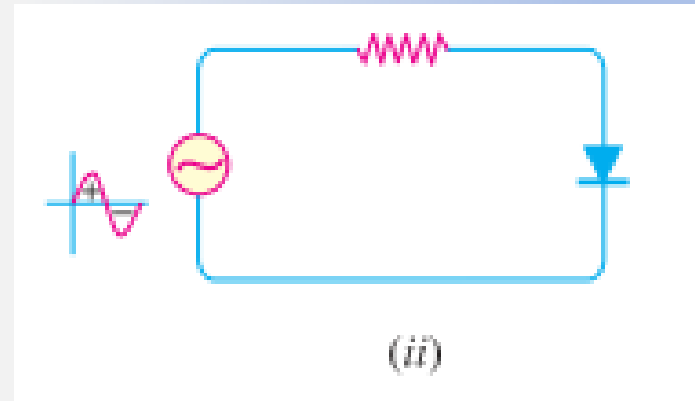
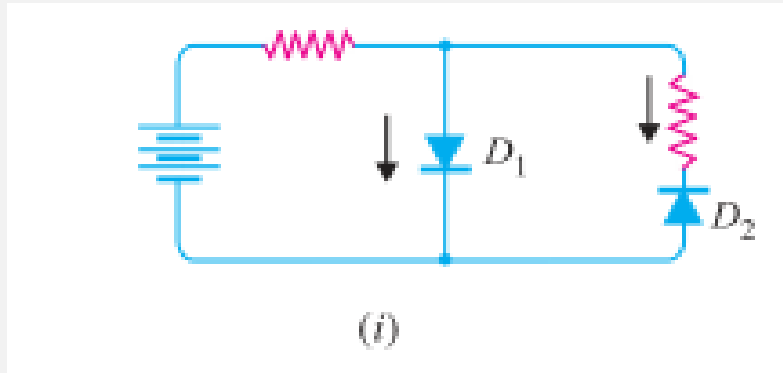
Reverse Current



- ❑ A small amount current is generated due to the minority carriers in p and n regions.
- ❑ These minority carriers are produced due to thermally generated hole-electron pairs.
- ❑ Minority electrons in p-region pushed towards +ve bias voltage, cross junction and then fall in the holes in n-region and still travel in valance band generating a hole current.

DIODE CONNECTION

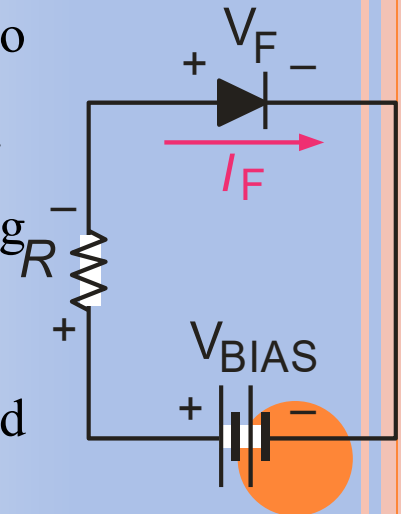
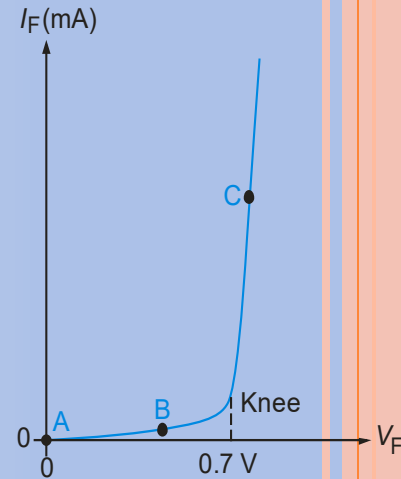
Determine the type of connection in each diode.



Diode V-I Characteristic

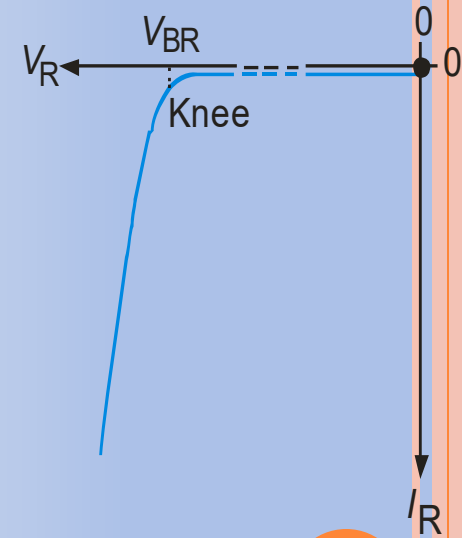
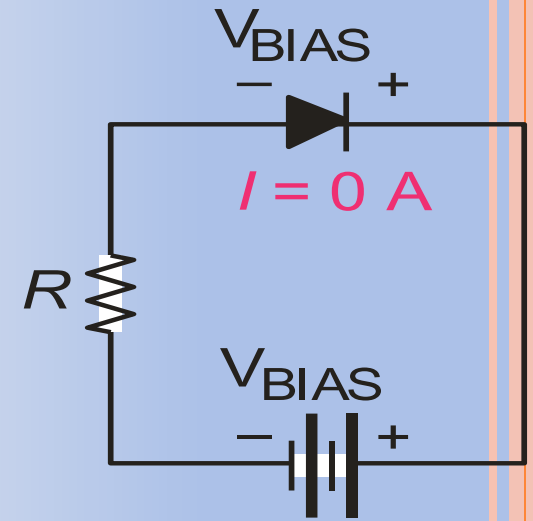
VI Characteristic for forward bias.

- ❑ The current in forward biased called *forward current* and is designated I_f .
- ❑ At 0V (V_{Bias}) across the diode, there is no forward current.
- ❑ With gradual increase of V_{Bias} , the forward voltage and forward current increases.
- ❑ A resistor in series will limit the forward current in order to protect the diode from overheating and permanent damage.
- ❑ A portion of forward-bias voltage drops across the limiting resistor.
- ❑ Continuing increase of V_f causes rapid increase of forward current but only a gradual increase in voltage across diode.



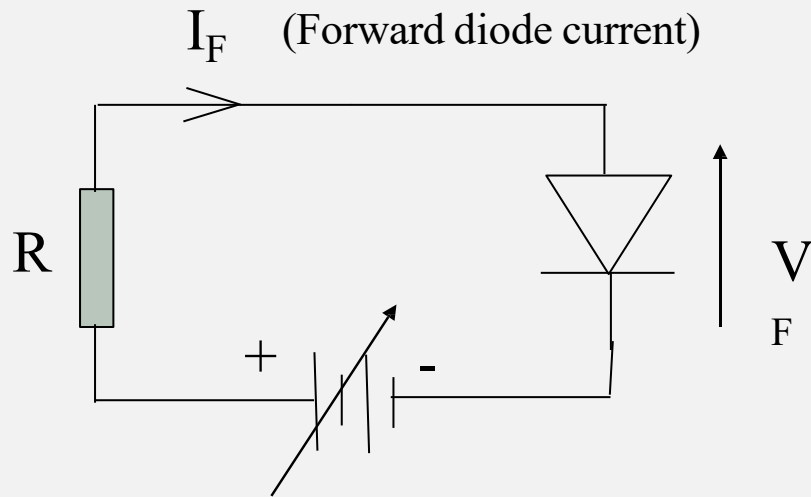
VI Characteristic for reverse bias.

- ❑ With 0V reverse voltage there is no reverse current.
- ❑ There is only a small current through the junction as the reverse voltage increases.
- ❑ At a point, reverse current shoots up with the break down of diode. The voltage called break down voltage. This is not normal mode of operation.
- ❑ After this point the reverse voltage remains at approximately V_{BR} but I_R increase very rapidly.
- ❑ Break down voltage depends on doping level, set by manufacturer.

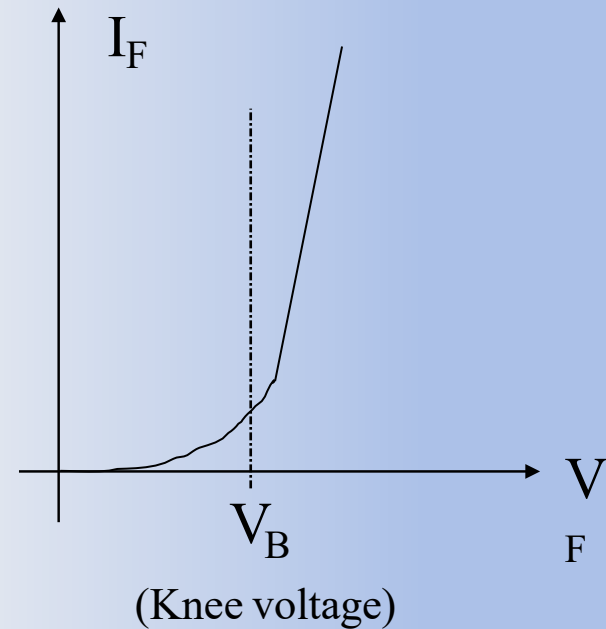


V-I CHARACTERISTIC OF JUNCTION DIODE

- Consider forward bias condition



$V_B \approx 0.7V$ (Si)
 $\approx 0.3V$ (Ge)



V-I CHARACTERISTIC OF JUNCTION DIODE

- The forward region I-V characteristic is approximated by exponential function:

$$I_F = I_S \left(e^{V_F / n V_T} - 1 \right)$$

Where:

- I_F = Diode forward current
- V_F = Diode forward voltage
- I_S = Saturation current
- V_T = Thermal voltage
- η is between 1 and 2 depending on the material and physical structure of diode

η (eta)

THE DIODE DATASHEET

- A manufacturer's datasheet gives detailed information on a device so that it can be used properly in a given application.
- A typical datasheet provides;
 - maximum ratings,
 - electrical characteristics,
 - mechanical data, and
 - graphs of various parameters.



THE DIODE DATASHEET

- The presentation of information on datasheets may vary from one manufacturer to another,
 - but they basically all convey the same information.
 - The mechanical information, such as package dimensions, are not shown in this lecture but are generally available from the manufacturer.
 - On this datasheet there are **three categories** of data given in table form and **four types of characteristics** shown in graphical form.
- 

FAIRCHILD
SEMICONDUCTOR®

Features

- Low forward voltage drop.
- High surge current capability.



DO-41
COLOR BAND DENOTES CATHODE

Absolute Maximum Ratings*

$T_a = 25^\circ\text{C}$ unless otherwise noted

Symbol	Parameter	Value							Units
		4001	4002	4003	4004	4005	4006	4007	
V_{RRM}	Peak Repetitive Reverse Voltage	50	100	200	400	600	800	1000	V
$I_{F(AV)}$	Average Rectified Forward Current, .375 " lead length @ $T_A = 75^{\circ}C$	1.0							A
I_{FSM}	Non-repetitive Peak Forward Surge Current 8.3 ms Single Half-Sine-Wave	30							A
T_{stg}	Storage Temperature Range	-55 to +175							$^{\circ}C$
T_J	Operating Junction Temperature	-55 to +175							$^{\circ}C$

* These ratings are limiting values above which the serviceability of any semiconductor device may be impaired.

Symbol	Parameter	Value	Units
P_D	Power Dissipation	3.0	W
$R_{\theta JA}$	Thermal Resistance, Junction to Ambient	50	$^{\circ}\text{C/W}$

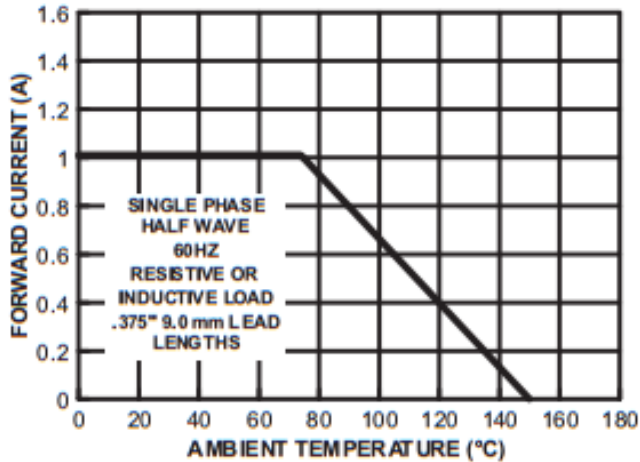
$T_A = 25^\circ\text{C}$ unless otherwise noted

Symbol	Parameter	Device							Units
		4001	4002	4003	4004	4005	4006	4007	
V_F	Forward Voltage @ 1.0 A	1.1							V
I_{rr}	Maximum Full Load Reverse Current, Full Cycle $T_A = 75^\circ\text{C}$	30							μA
I_R	Reverse Current @ rated V_R $T_A = 25^\circ\text{C}$ $T_A = 100^\circ\text{C}$	5.0 500							μA μA
C_T	Total Capacitance $V_R = 4.0\text{ V}$, $f = 1.0\text{ MHz}$	15							pF

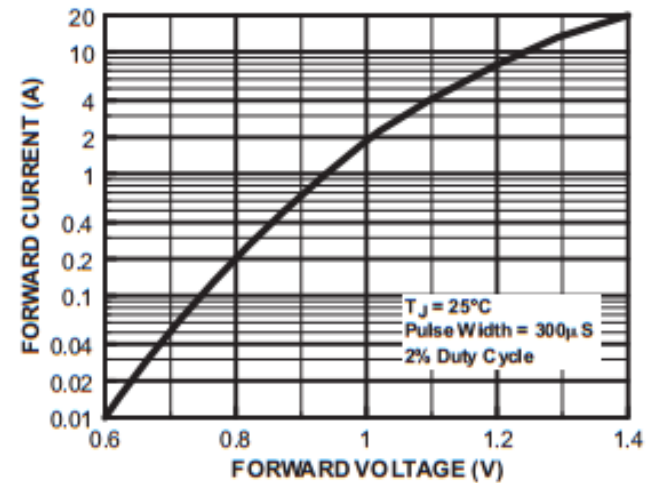
THE DIODE DATASHEET

Typical Characteristics

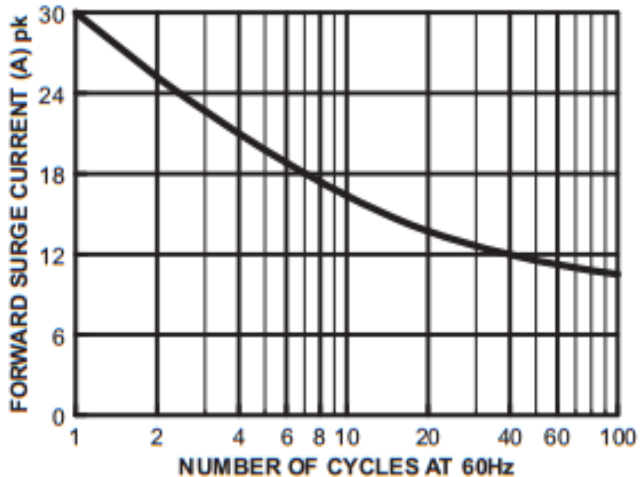
Forward Current Derating Curve



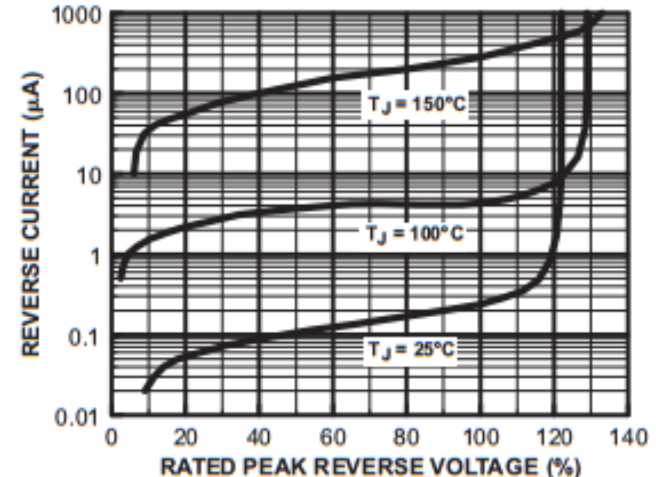
Forward Characteristics



Non-Repetitive Surge Current



Reverse Characteristics



Data Categories

- *Absolute Maximum Ratings*: Maximum values of the several parameters under which the diode can be operated without damage or degradation at ambient temperature
- Ambient temperature is the temperature of the air surrounding the device - 25°C , unless otherwise stated.
- For greatest reliability and longer life, the diode should be operated well under these maximums.



Data Categories

- *Thermal Characteristics*: All devices have a limit on the amount of heat that they can tolerate without failing in some way.
- Average power dissipation is the amount of power that the diode can dissipate under any condition.
- A diode should never be operated at maximum power, except for brief periods, to assure reliability and longer life.



Data Categories

- *Electrical Characteristics:* The electrical characteristics are specified under certain conditions and are the same for each type of diode.
- These values are typical and can be more or less for a given diode.
- Some datasheets provide a minimum and a maximum value in addition to a typical value for a parameter.

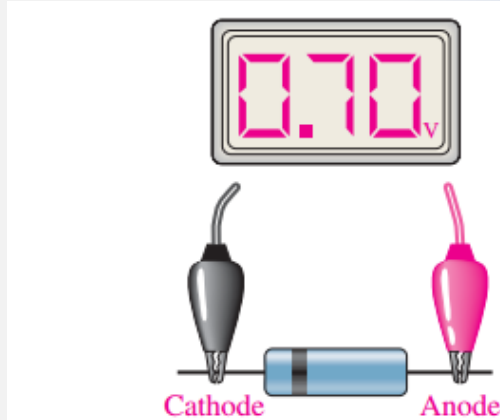
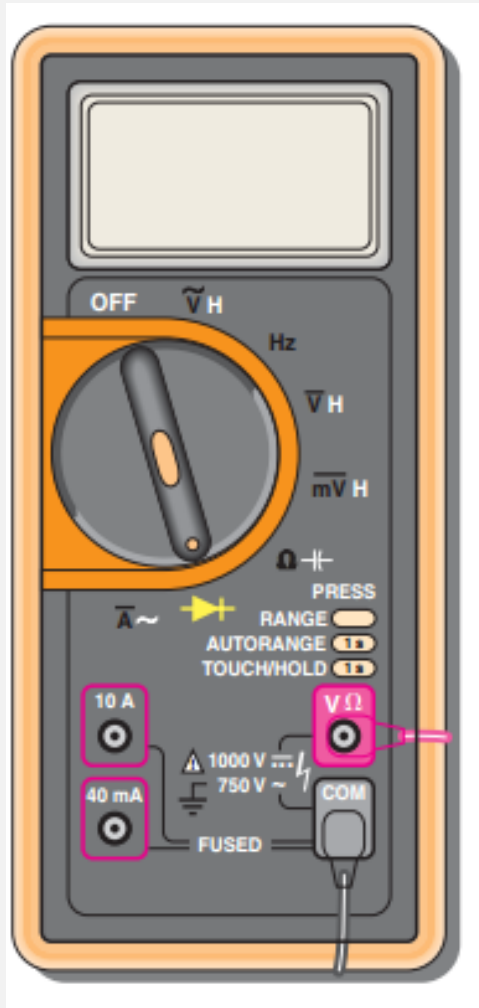


Testing a Diode

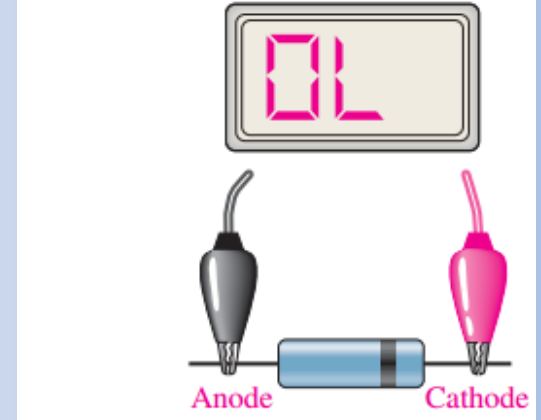
- A good diode:
 - **reverse bias:** extremely high resistance (ideally an open)
 - **forward bias:** A very low resistance
- A defective diode:
 - **Open diode :** an extremely high resistance (or open) for both forward and reverse bias
 - **Shorted or resistive diode:** zero or a low resistance for both forward and reverse bias.
- *An open diode* is the most common type of failure.



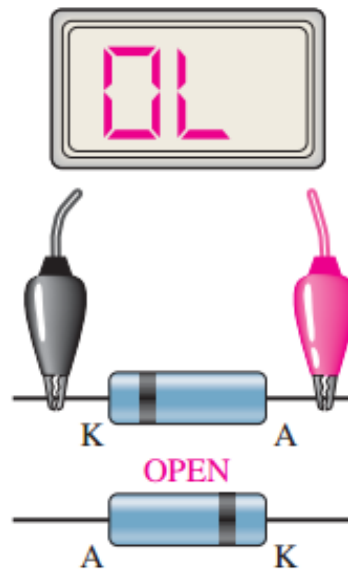
Testing a Diode



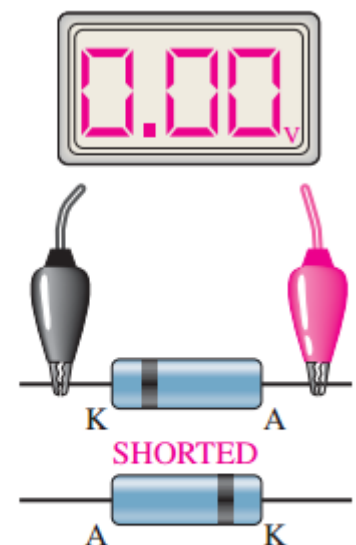
(a) Forward-bias test



(b) Reverse-bias test



(c) Forward- and reverse-bias tests for an open diode give the same indication.



(d) Forward- and reverse-bias tests for a shorted diode give the same 0 V reading.

NEXT SESSION ...TRANSISTORS (BJT)

Thank you for listening

